How Many Labels Does Eye-Contact Detection Need?

A Subject-Independent Comparison of Geometric Features, Visual Foundation Models, and Gaze Estimators, and Where Gaze Lives Inside Foundation Models

FewGaze authors

Code, configurations, result files and a live demo: `fewgaze` repository

Abstract

How many labelled images does a model need to recognise that a person it has never seen is looking into the camera? We study this question on the Columbia Gaze Data Set (5,878 images of 56 people after an audit removed a duplicated pair with conflicting annotations) under a protocol designed to rule out leakage: identity-disjoint folds, one shared set of training images per budget ($k \in \{5, 10, 25, 50, 100, \text{ALL}\}$ per class, five seeds), a subject-level bootstrap, and an analysis plan committed to version control before the first experiment. *Study 1* compares four frozen representations with a fixed logistic-regression head. Nine MediaPipe eye and head measurements with a quadratic boundary are the most label-efficient at every budget, reaching AUROC 0.537 at k=5, 0.730 at k=100 and 0.810 with all data. A DINOv2 linear probe stays at chance ($\leq$ 0.513). The L2CS-Net gaze estimator scores 0.595 without any labels, but a head trained on its two angles never improves on this, because its outputs carry a head-pose-dependent bias. A 5° gaze shift moves the iris by only 0.025 eye-widths, against a between-person spread of 0.017, which explains why no method separates the gaze-locked condition from its nearest neighbours. *Study 2* (pre-registered) probes every layer of five frozen foundation models (DINOv2-S/B, CLIP, SigLIP, MAE) from the summary token and from eye-region patch tokens. [Study 2 summary sentence, filled from results]. A browser demo reproduces the Python pipeline to within 4×10^{-4} in probability.

1 Introduction

Knowing when someone looks at a camera matters for video-mediated communication, attentive user interfaces and social robots. Prior work detects eye contact with a camera [21], with arbitrary objects without supervision [27], with mobile devices in everyday life [3] and between people in group conversations [11, 12]. Each new setting needs labelled examples, and the value of each label depends on the representation that consumes it. Generic foundation models such as DINOv2 [13] and CLIP [18] are often expected to be sample-efficient. Dedicated gaze estimators encode gaze explicitly [1, 4]. Hand-crafted eye geometry is cheap and interpretable [2, 7]. How these compare when labels are scarce, and when evaluation is on *new people*, has rarely been measured directly.

We study a deliberately narrow target: **camera-directed gaze under controlled capture**, the "gaze locking" condition of Smith et al. [21]. It is a proxy for eye contact, not interpersonal eye contact. Our contributions are:

1. a leakage-tested, subject-independent label-budget protocol in which every method shares folds, training images and test images, with the plan committed before the first run and an exact rerun from a clean commit;
2. learning curves, paired comparisons, label-noise, head-pose and per-person analyses for four representation families (Study 1);
3. a mechanistic account of the failures: the size of the 5° signal relative to person-to-person variation, identity-dominated foundation-model embeddings, and a head-pose-dependent bias in a pretrained gaze estimator;
4. a pre-registered, layer- and token-resolved probe of five frozen foundation models for gaze under held-out-subject evaluation, linked to eye-contact classification (Study 2);
5. an open, parity-tested in-browser demonstrator.

2 Related work

Eye contact and gaze locking. Smith, Yin, Feiner and Nayar introduced *gaze locking*, passive detection of camera-directed gaze, together with the Columbia Gaze Data Set; they classified rectified eye patches with PCA, MDA and an SVM [21]. Zhang, Sugano and Bulling detected eye contact with arbitrary targets by discovering gaze targets without supervision [27]; Sugano, Zhang and Bulling aggregated many viewers' gaze to estimate audience attention on public displays [23]. Bâce et al. quantified everyday attention to mobile devices with eye-contact detection on front-camera video [3]. Müller, Huang, Zhang and Bulling combined gaze and speaking behaviour to detect eye contact in multi-person conversation [11], Müller et al. detected low rapport in small groups from non-verbal behaviour including gaze [10], and the MultiMediate challenge established eye-contact detection as a benchmark task [12]. None of these works measures sample efficiency across representation families under a shared protocol.

Appearance-based gaze estimation. Sugano et al. trained person-independent 3D gaze estimators from synthesised eye images [22]. Zhang et al. introduced MPIIGaze for gaze estimation in the wild [26, 31], showed that full-face input with spatial weights improves accuracy [28], revisited the role of data normalisation [29], compared appearance-based estimators with commercial eye trackers and discussed implications for applications [30], and released ETH-XGaze for extreme head poses [32]. Gaze360 provides unconstrained 360° annotations [8], on which L2CS-Net trains per-angle binned classifiers [1]. Park et al. estimated gaze from learned

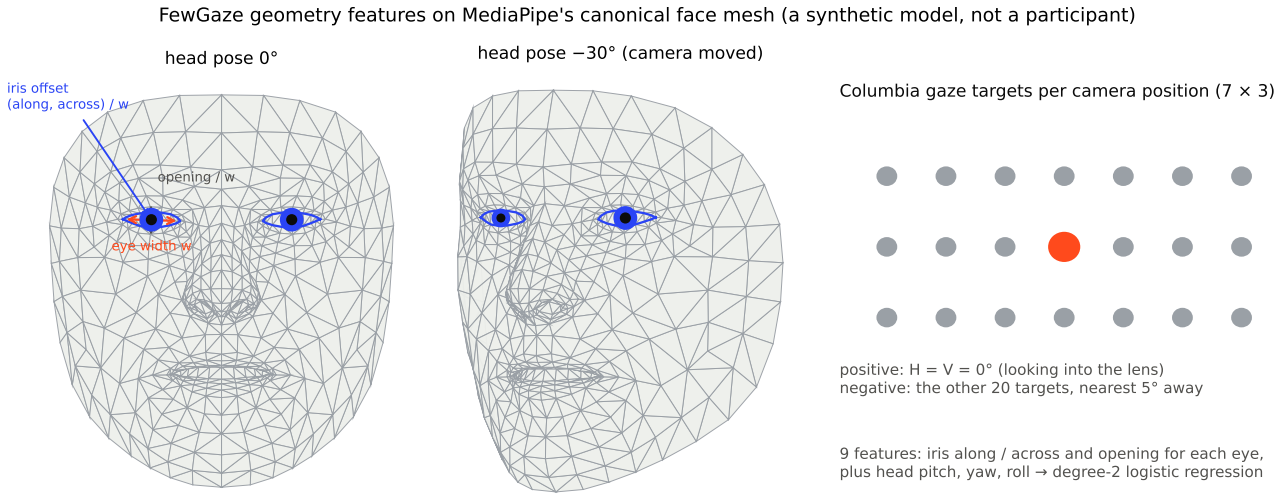


Figure 1: What is measured. The nine geometry features drawn on MediaPipe's canonical face mesh (a synthetic model; no participant is shown). Each eye contributes the iris offset along and across the corner-to-corner axis and the eyelid opening, all divided by eye width; three head-rotation angles complete the vector. Right: Columbia's 7 × 3 gaze targets per camera position; only the central target is positive.

eye-region landmarks [14] and, in FAZE, adapted a person-specific estimator from as few as three calibration samples [15]. Cheng et al. survey and benchmark the field [4]. Our geometry baseline uses MediaPipe face-mesh and iris landmarks [2, 7, 9].

Foundation models and gaze. Self-supervised and image–text models (DINOv2 [13], CLIP [18], SigLIP [25], MAE [6]) are widely used as frozen backbones. El Banani et al. probed such models for 3D awareness [5]. For gaze, UniGaze argues that general-purpose self-supervised models designed for semantic tasks transfer poorly and pre-trains MAE-style on faces instead [17]; CLIP has been used to improve gaze generalisation through language [24]; and a frozen DINOv2 encoder supports gaze-*target* estimation [19]. Closest to Study 2, a recent preprint probes frozen DINOv2, CLIP and SigLIP features for gaze on MPIIFaceGaze with a random image split [20]. We add held-out-subject evaluation, every layer, a contrast between the summary token and eye-region patches, and the link to eye-contact classification.

3 Data and protocol

Dataset and label. In Columbia Gaze [21], 56 people sat in a chin rest 2 m from a camera that moved between five positions (head poses 0°, ±15°, ±30° relative to the camera) and fixated a 7 × 3 grid of targets defined relative to that camera ($H \in \{0, \pm5, \pm10, \pm15\}°$, $V \in \{0, \pm10\}°$; Fig. 1). Positives are H=V=0; the other 20 cells are negatives. Our audit confirmed 5,880 readable 5184×3456 images in a complete grid, and found one byte-identical pair with conflicting horizontal-gaze annotations (subject 0007); both images were excluded before modelling, leaving 280 positives and 5,598 negatives (1 : 20). Fig. 2 summarises the composition.

Splits, budgets and selections. Subjects are dealt into five folds with a fixed seed (11–12 test people per fold); automated tests verify disjointness, coverage and determinism. For each fold and seed $s \in \{0..4\}$, one permutation per class of the

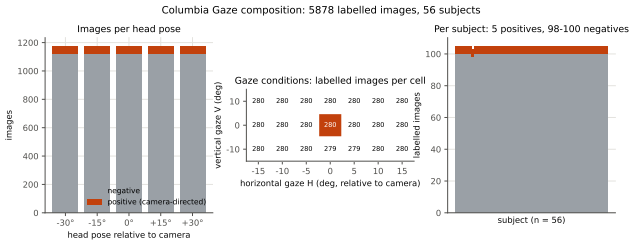


Figure 2: Dataset after the audit: images per head pose, labelled images per gaze cell, and per-person class counts.

training people's images is drawn and its first k images are used. Budgets are therefore nested, and **every method receives the same images**. ALL is every training-person image (about 225 positives and 4,480 negatives).

Representations (frozen) and classifier. *Geometry* uses nine MediaPipe features (Fig. 1). *DINOv2-S* [13] uses the CLS token of a 448 px face crop resized to 224^2. *L2CS-Net* [1] (ResNet-50, Gaze360 weights) gives (pitch, yaw) from the 448^2 crop; we also score it without labels as minus the angle to the camera axis. *CLIP* ViT-B/32 [18] is scored zero-shot with two prompts fixed in advance. Every trained method uses scikit-learn [16] logistic regression (C=1, balanced weights, threshold 0.5) inside a pipeline fit on training people only; hyperparameters were never tuned. Because the positive class is the *centre* of gaze space, no linear function of a monotone gaze proxy can isolate it; a degree-2 expansion was therefore prespecified for the low-dimensional geometry and L2CS features, and embeddings get a linear probe.

Metrics and uncertainty. AUROC (primary) and balanced accuracy are averaged over 25 fold×seed runs. 95% intervals come from a fold-stratified *subject* cluster bootstrap (1,000 replicates) computed in closed form (AUROC = $c^\top A c / [(c \cdot P)(c \cdot N)]$ for subject counts c; tested against explicit resampling to 10^{-12}). The same resamples are used for every method, so differences are paired. Intervals reflect

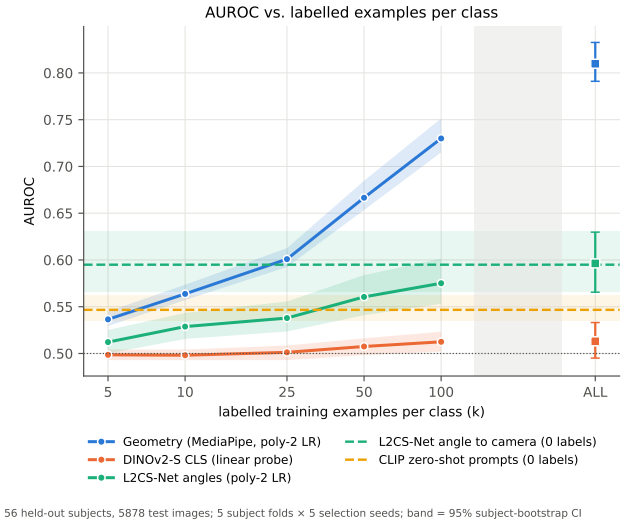


Figure 3: AUROC on held-out people versus labelled examples per class (log scale); bands are 95% subject-bootstrap intervals, dashed lines use no FewGaze labels.

Method	$k=5$	$k=10$	$k=25$	$k=50$	$k=100$	ALL
Geometry (MediaPipe, poly-2 LR)	0.537	0.564	0.601	0.667	0.730	0.810
DINOv2-S CLS (linear probe)	0.499	0.498	0.501	0.507	0.512	0.513
L2CS-Net angles (poly-2 LR)	0.512	0.529	0.538	0.560	0.575	0.596
Geometry (linear LR)	0.504	0.509	0.515	0.516	0.520	0.555
DINOv2-S mean-patch (linear probe)	0.499	0.500	0.502	0.510	0.521	0.545
L2CS-Net 2048-d features (linear probe)	0.513	0.530	0.562	0.590	0.615	0.633
CLIP ViT-B/32 (linear probe)	0.505	0.507	0.516	0.529	0.545	0.567
L2CS-Net angle to camera (0 labels)	0.595 (no training labels)					
CLIP zero-shot prompts (0 labels)	0.547 (no training labels)					

Table 1: Mean AUROC over 5 folds × 5 seeds. Top: primary methods; middle: ablations; bottom: no labels.

which people were tested, not retraining variability, which we report separately as the SD across runs.

4 Study 1: how many labels?

Learning curves. Geometry led at every budget (Fig. 3, Table 1): AUROC 0.537 [0.529, 0.545] at k=5, 0.601 [0.592, 0.613] at 25, 0.730 [0.715, 0.751] at 100 and 0.810 [0.791, 0.833] with all data (balanced accuracy 0.527, 0.578, 0.686, 0.733). Its curve was also the steepest, gaining 0.19 AUROC between 5 and 100 labels. All three prespecified paired comparisons excluded zero at every budget (Fig. 4); for L2CS−DINOv2 at k=5 the lower bound is only just above zero. The DINOv2 linear probe stayed at chance (0.499–0.513). L2CS angles with a trained head rose from 0.512 to 0.596, matching but not improving on its zero-label score (0.595 [0.566, 0.630]). CLIP zero-shot reached 0.547 [0.535, 0.563]. Geometry needed between 10 and 25 labels per class to overtake the zero-label gaze estimator.

Ablations. Removing the polynomial left geometry near chance (0.504–0.555), confirming the centre-of-gaze-space argument. L2CS's 2048-d penultimate features with a linear probe (0.633 at ALL) beat its two output angles, and neither DINOv2 mean-patch tokens (≤ 0.545) nor a CLIP probe (≤ 0.567) approached geometry (Fig. 5).

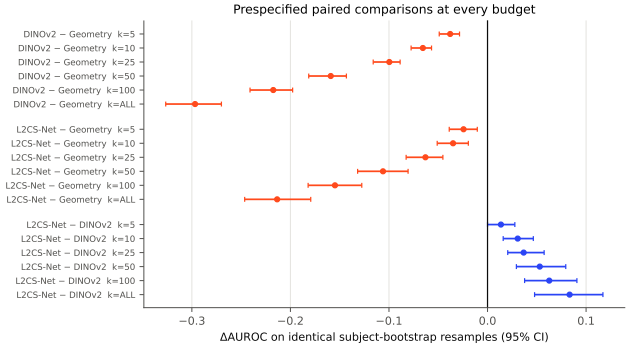


Figure 4: Prespecified paired differences (ΔAUROC) at every budget, on identical bootstrap resamples.

The 5° problem. At k=25, geometry called 71.8% of true positives camera-directed, but also 67.7% of the $|H|$=5° negatives, 50.8% of $|H|\geq$10° negatives and 43.2% of the $V=\pm$ 10° negatives (Fig. 7). The reason is visible in the raw measurements (Fig. 6): moving the gaze from 0° to +5° shifts the median iris by 0.025 eye-widths, while people already differ by 0.017 (interquartile range) at 0°. This is consistent with the ±2.5° horizontal tolerance that Smith et al. estimated for their own detector [21].

Head pose and label noise. With all data, geometry's AUROC fell from 0.891 at a frontal pose to 0.835/0.847 at ∓15° and 0.718/0.766 at ∓30° (Fig. 8). Flipping training labels degraded geometry roughly in proportion to the flip rate (at k=100: 0.730 → 0.695/0.656/0.602 for 10/20/30%; Fig. 9); at k=25, 10% noise (0.582) cost less than reducing the clean budget to k=10 (0.564).

Why the gaze estimator saturates. Fig. 10 plots L2CS-Net's raw outputs. Yaw tracks horizontal gaze, but at the gaze-locked target its mean ranges from +1.6° (head −30°) to −15° (head +30°): part of what it reports as gaze is head pose. Pitch tracks vertical gaze with a constant offset of roughly −6° to −9°. Per person, the zero-label score ranges from AUROC 0.32 to 0.94 (SD 0.14, versus 0.05 for geometry; Fig. 11), the signature of person-specific calibration error, which a pooled two-feature classifier cannot remove and which per-person adaptation in the style of FAZE [15] targets. The two methods fail on different people (Fig. 11), which suggests, but does not establish, complementarity.

Why the foundation model fails. A t-SNE of the embeddings (Fig. 12) shows DINOv2 collapsing each person into one tight cluster, split further by head pose: the nearest neighbour of an image is the same person in > 99.9% of cases (86% for geometry). A ridge decoder recovers head pose from the final CLS token with held-out-subject R^2=0.95 but not gaze (R^2<0). Feeding DINOv2 an eye-region crop instead (a post-hoc analysis) raised AUROC at ALL only from 0.513 to 0.562. Study 2 asks whether gaze is present anywhere inside such models.

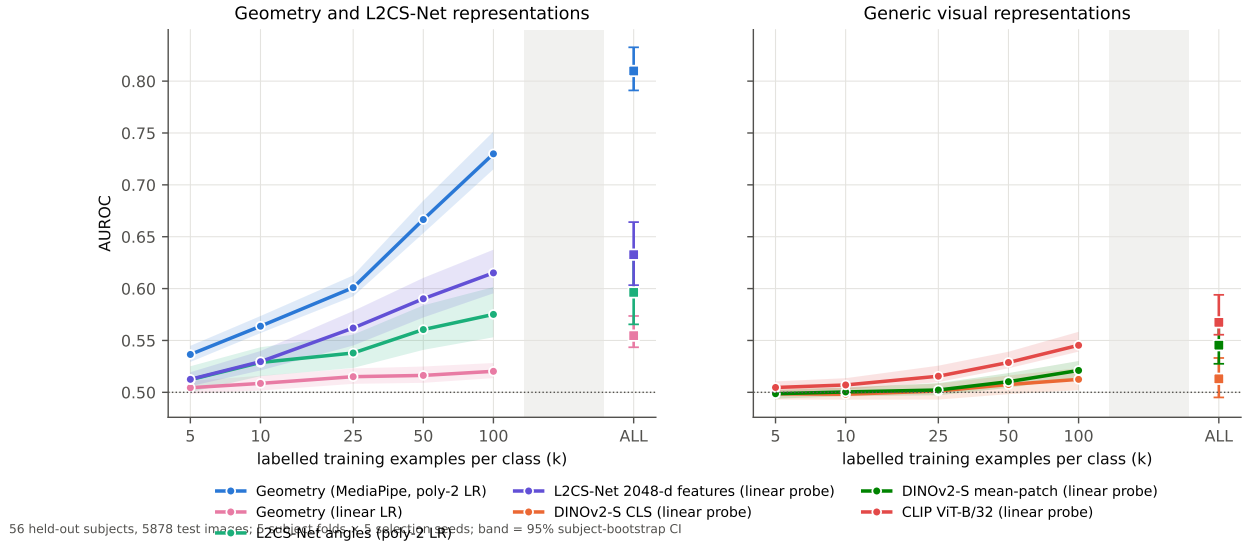


Figure 5: Ablations: representation choices within each family.



Figure 6: The 5° problem. Mean iris offset per person (dots, n=56 per cell) at frontal head pose. The gaze-locked cell (red) overlaps heavily with its ±5° neighbours.

5 Study 2: where does gaze live inside foundation models?

Design (pre-registered). We froze DINOv2-S/14, DINOv2-B/14 [13], CLIP ViT-B/16 [18], SigLIP-B/16 [25] and MAE ViT-B/16 [6] (MAE with no masking and identity patch order), fed each the same 224^2 face crop with its own normalisation, and cached two representations at every one of 13 layers: the summary token (CLS; SigLIP has none, so the mean patch) and the mean of the patch tokens over each eye (eye boxes from MediaPipe landmarks re-detected on the crop, padded by 25% of eye width). A ridge decoder (α by internal cross-validation) predicted H, V and head pose on held-out people, using the same five folds; a linear logistic probe and a geometric readout $-(\hat{H}/\sigma_H)^2 - (\hat{V}/\sigma_V)^2$ scored eye contact. The readout uses continuous gaze labels, which is more supervision than Study 1's binary labels. The nine geometry features were probed identically as a reference.

[Study 2 results paragraphs, figures fig_s2_layers, fig_s2_heat, fig_s2_readout and table: filled from results/study2.]

6 An in-browser demonstrator

We exported the geometry classifier trained on all 5,878 images and re-implemented the feature code in JavaScript on MediaPipe tasks-vision 0.10.21 (the same face-landmark model file as the Python pipeline). On 24 images resized exactly as in Python, the browser matched Python to 5.5×10^{-6} in landmark coordinates, 2×10^{-3} in features and 4.3×10^{-4} in probability, with no decision flips; the test also revealed that the library's transformation matrices are column-major, a detail whose mistreatment flipped 13 of 24 decisions. The demonstrator runs on camera frames or photos entirely on the client, flags inputs outside the training ranges, and states that it is a research artefact, not a detector.

7 Discussion

Under this protocol the most label-efficient representation was the simplest: nine geometric measurements with the right inductive bias for a target that sits at the centre of gaze space. Generic self-supervised embeddings were not label-efficient here because their dominant variation is identity and head pose, both irrelevant to the task. The gaze estimator shows

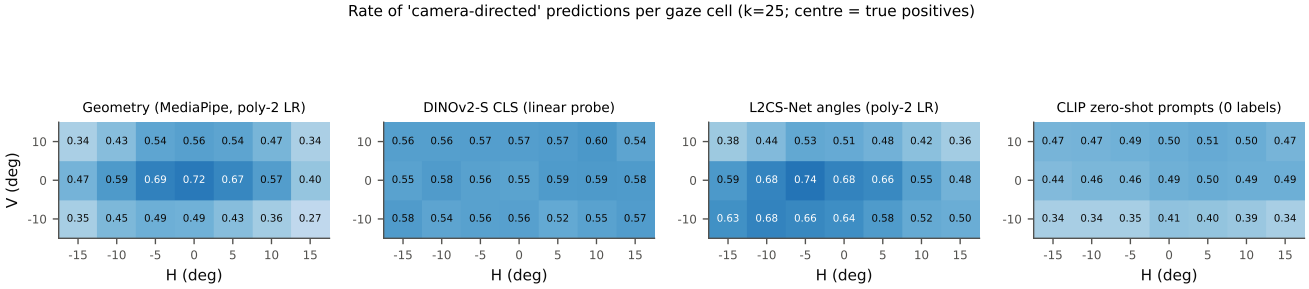


Figure 7: Share of images predicted camera-directed per gaze cell at k=25 (centre = true positives).

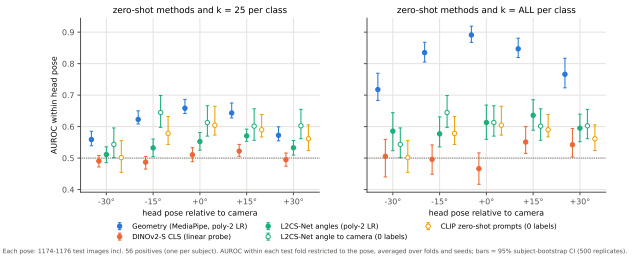


Figure 8: AUROC within each head pose (56 positives per pose); hollow markers use no labels.

the value and the limits of prior knowledge: it beats every method trained on ten labels or fewer without seeing a single FewGaze label, but its person- and pose-dependent bias caps what a pooled classifier can learn from its outputs. [Study 2 discussion sentence.] No method resolved the dataset's hardest distinction, 0° versus 5°, and absolute performance (AUROC $\leq$ 0.81) is far from deployable; it is not comparable to the per-image accuracy reported by Smith et al., who evaluated a different detector under a different protocol [21].

8 Limitations and ethics

Results hold for one controlled dataset: 56 people, one lab, 2 m, yaw-only head pose, a chin rest. Camera-directed gaze is a proxy, not interpersonal eye contact, and individual fixation accuracy is unknown. Intervals capture subject sampling with models held fixed. Heads were fixed in advance; other probes might rank the embeddings differently. The dataset has no demographic labels, so we make no fairness claims, and pretrained models carry the biases of their training data. The dataset licence forbids redistribution, so no participant image appears in this paper. Gaze sensing can be misused for surveillance or attention scoring; our models are research artefacts and are unsuitable for monitoring or high-stakes use.

9 Conclusion

A handful of interpretable measurements with a quadratic boundary made the best use of labels, from 5 per class to several thousand; a frozen generic embedding with a linear probe did not learn the task, and a pretrained gaze estimator mainly helped when no labels were available. [Study 2 conclusion.] The open question is whether person-specific calibration or eye-local representations can resolve the last 5° that separate looking *at* the camera from looking just past it.

Reproducibility. Every number, table and figure is regenerated by the commands in the repository from the recorded runs; the plans for both studies were committed before they ran; images are not redistributed.

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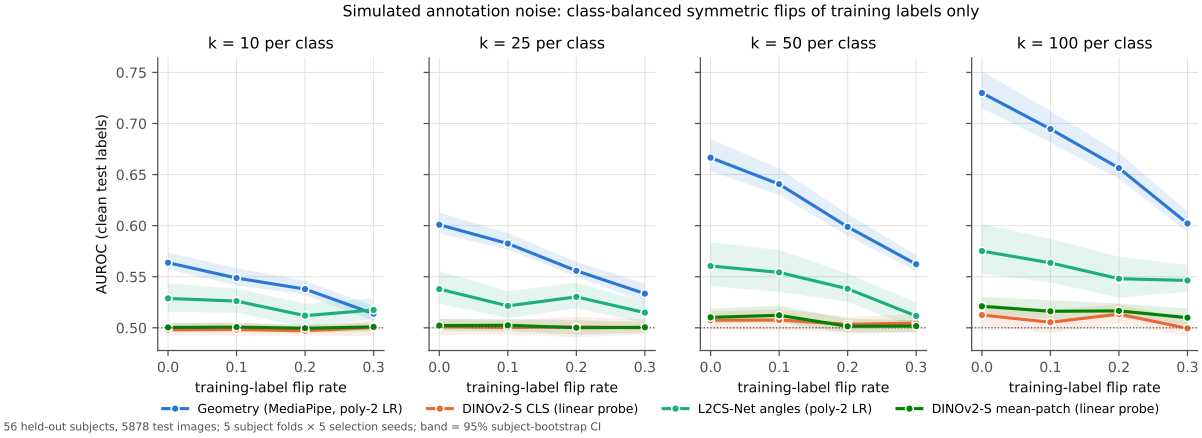


Figure 9: Simulated label noise: class-balanced flips of training labels only, identical across methods.

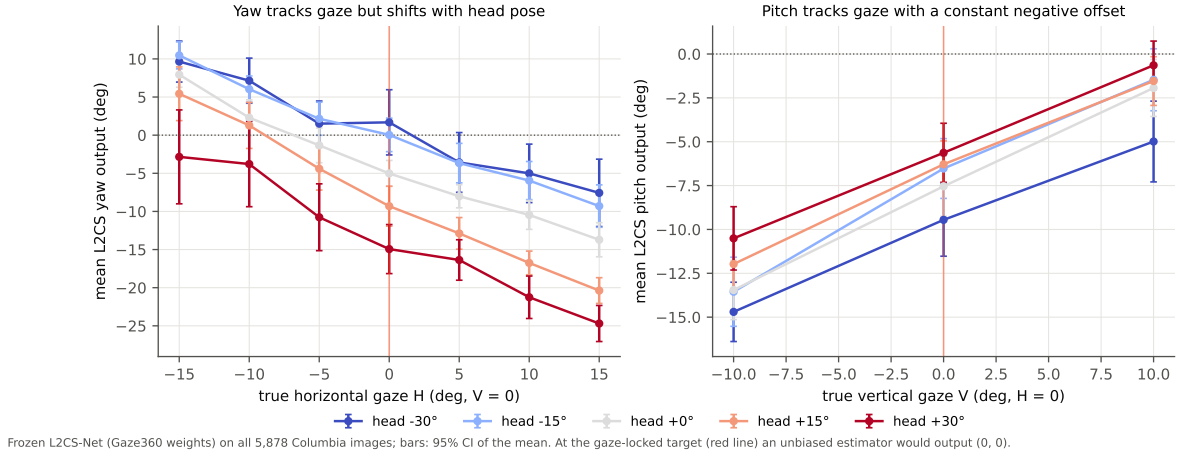


Figure 10: L2CS-Net is biased by head pose. Mean frozen L2CS outputs on Columbia versus true gaze, per head pose. An unbiased estimator would pass through (0, 0) at the red line for every pose.

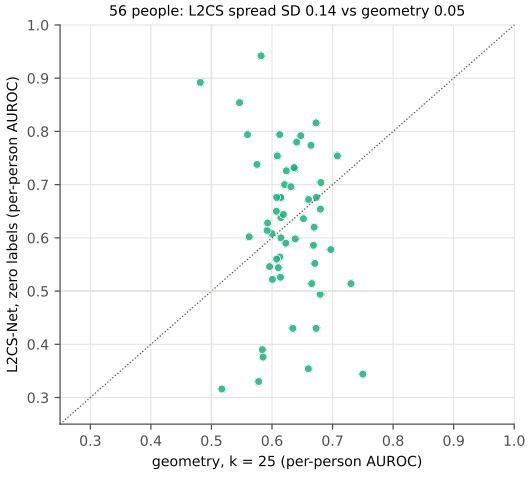


Figure 11: Per-person AUROC: L2CS-Net without labels versus geometry at k=25.

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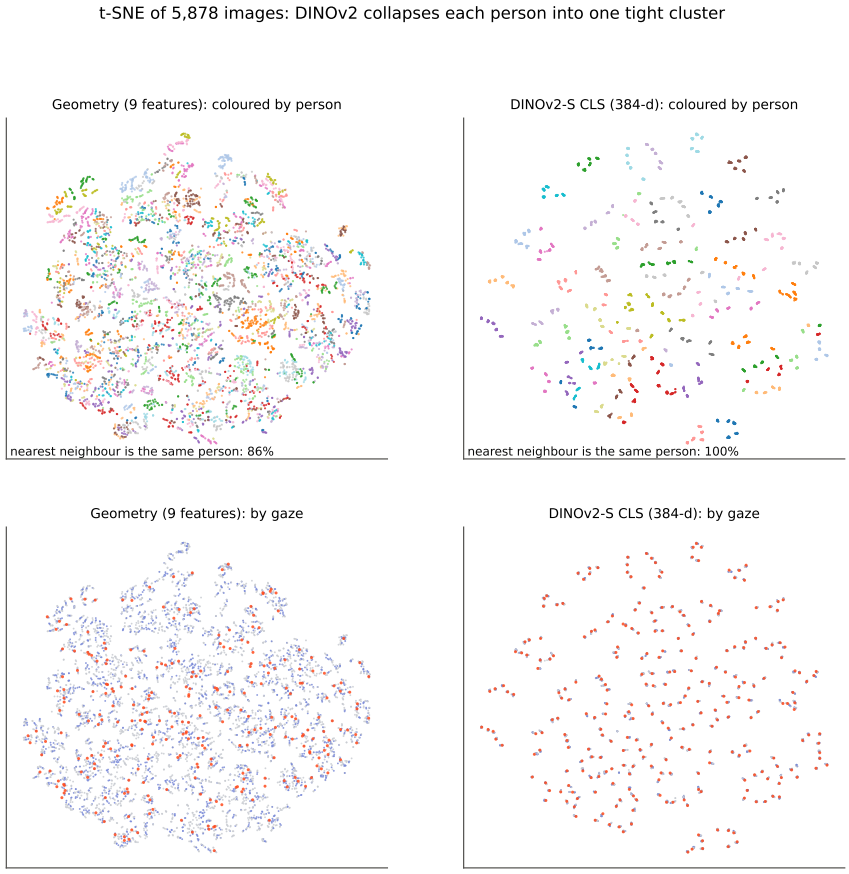


Figure 12: *t*-SNE of 5,878 images. Top: coloured by person; bottom: by gaze condition.

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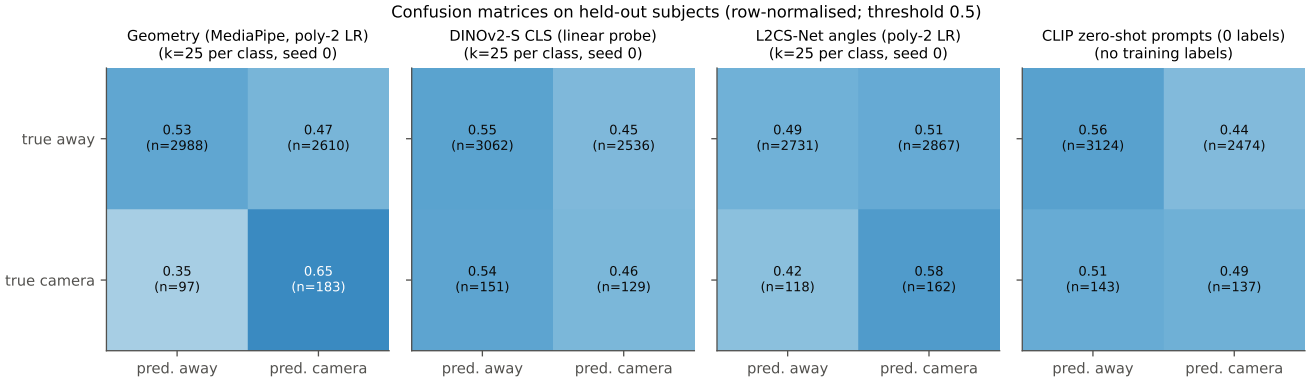


Figure 13: Confusion matrices on held-out people (row-normalised), k=25, seed 0; CLIP uses no labels.

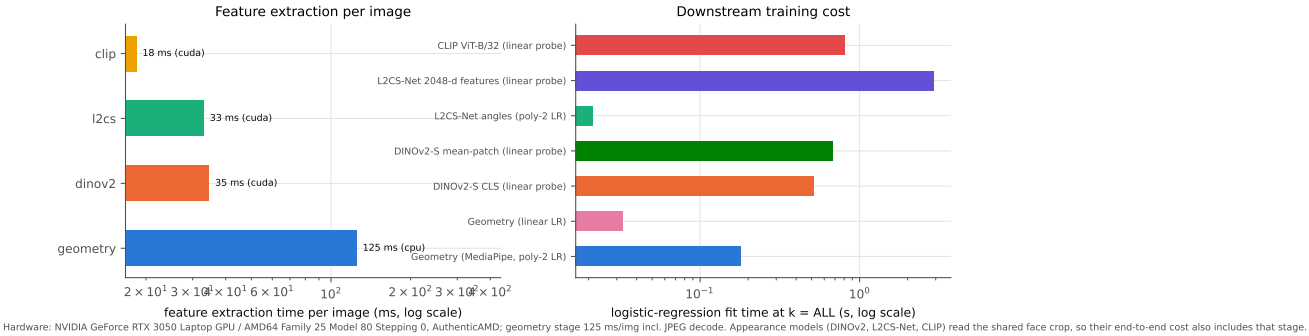


Figure 14: Feature-extraction and training cost on one laptop (RTX 3050 laptop GPU).

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